MASTER FUNCTIONAL SKILLS (MFS) · WORKED SOLUTIONS ·
MasterFunctionalSkills.co.uk · AQA Maths Level 2 Specialist
Full Marks (100% Pass) · Verified by Tutor Abdul · January 2023 - Paper 2

AQA

Please write clearly in block capitals.

Centre number M F S 0 1 Candidate number 2 0 2 6

Surname MASTER

Forename(s) FUNCTIONAL SKILLS

Candidate signature Master Functional Skills

I declare this is my own work.

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Candidate: Master Functional Skills
Date: 20/10/2026 · Tutor Abdul

# Functional Skills Level 2 MATHEMATICS

## Paper 2 Calculator

Monday 9 January 2023 Afternoon Time allowed: 1 hour 30 minutes

**Materials**

For this paper you must have:
- a calculator
- mathematical instruments.



| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1–6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| TOTAL | |

**Instructions**
- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

**Information**
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- If your calculator does not have a π button, take the value of π to be 3.142

**Advice**

In all calculations, show clearly how you work out your answer.

J A N 2 3 8 3 6 2 2 0 1

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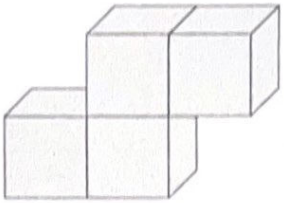
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## Section A

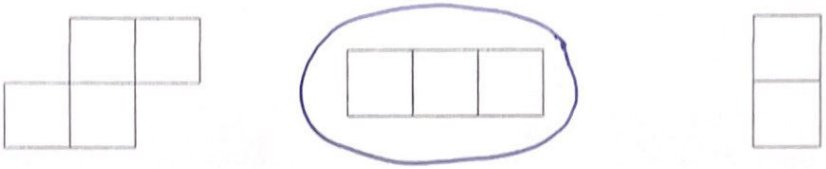
Answer **all** questions in the spaces provided.

**1** Here is a solid made from four cubes.



Circle the plan view of the solid.

**[1 mark]**



**2** Write the number 2 307 049 in words.

**[1 mark]**

Two million, three hundred and seven thousand and forty nine.

Do not write outside the box

**3** Which is closer in value to 0.5

0.379 or 0.614 ?

You **must** show your working.

**[2 marks]**

$0.614 - 0.5 = 0.114$

$0.5 - 0.379 = 0.121$

$0.114 < 0.121$ so 0.614 is closer in value to 0.5

Answer 0.614

**4**

Surface area of a sphere = $4\pi r^2$

Work out the surface area of a sphere with a radius of 2.3 cm
State the units of your answer.

**[3 marks]**

$4 \times \pi \times 2.3^2 = 4 \times \pi \times 2.3 \times 2.3$

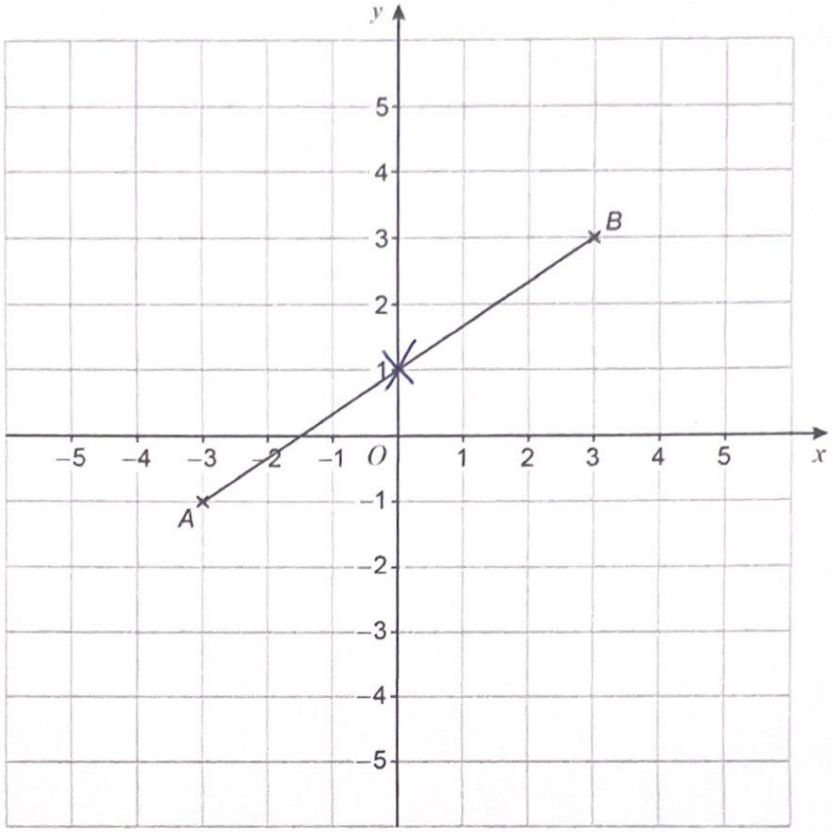
$= 21.16\pi$

$= 66.48$

Answer $21.16\pi$ or $66.48\ cm^2$

Do not write outside the box

**5**




Point *C* is the midpoint of line *AB*.

Plot point *C* on the grid and write down the coordinates of *C*.

**[2 marks]**

see graph

Answer ( 0 , 1 )

Do not write outside the box

6 The table contains grouped data.

| $x$ | Frequency | Mid-point | |
|---|---|---|---|
| $0 < x \leqslant 20$ | 7 | 10 | 70 |
| $20 < x \leqslant 40$ | 9 | 30 | 270 |
| $40 < x \leqslant 50$ | 4 | 45 | 180 |
| | **Total = 20** | | 520 |

Work out an estimate of the mean of $x$.

**[3 marks]**

See table

7 x 10 = 70

9 x 30 = 270

4 x 45 = 180

70 + 270 + 180 = 520

520 ÷ 20 = 26

Answer 26

12

**Turn over for Section B**

Do not write outside the box

## Section B

Answer **all** questions in the spaces provided.

**7 School office**

Ashley works in a school office.

**7 (a)** Ashley needs 29 book bags.

He sees these offers online.

| **Best book bags** | **Bags and stuff** |
|---|---|
| Book bags £4.99 each<br>For every 7 you buy you get an extra 1 free<br>Free delivery | Book bags reduced to £4.25 each<br>Delivery charge £3.90 per order |

Ashley wants the cheapest total price for the 29 bags and delivery.

Which offer should he choose?
You **must** show your working.

**[5 marks]**

Best Book Bags

29 ÷ (7+1) = 3.625 (3 free)

29 - 3 = 26 26 x 4.99 = 129.74

Bags and stuff

29 x 4.25 + 3.9 = 127.15

127.15 < 129.74

so should choose Bags and Stuff.

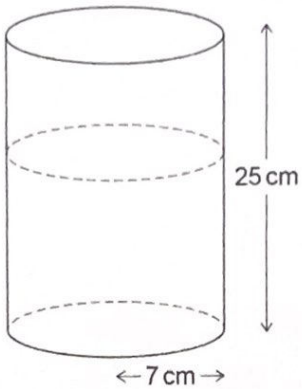
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**7 (b)** Ashley needs to put drinking water into 40 jugs before break time.

Each jug

is a cylinder with radius 7 cm and height 25 cm

will be $\frac{3}{5}$ full.




Water flows out of the tap at 13 litres per minute.

$1000\,cm^3 = 1$ litre

Can Ashley put the right amount of water into all 40 jugs in 6 minutes?
You **must** show your working.

**[6 marks]**

Volume $= \pi r^2 L = \pi \times 7 \times 7 \times 25$

$= 1225\pi$

$3/5 \times 1225\pi = 735\pi = 2309.1$

$40 \times 2309.1 \div 1000 = 92.36...$ how many litres she needs to fill

$13 \times 6 = 78$ Litres per minute $< 92.36...$

No she does not have time

11

Do not write outside the box

**8 Lifeguard**

Talia is a lifeguard at a swimming pool.

**8 (a)** One morning, Talia checked the water temperature in the pool each hour.

| Time | 8 am | 9 am | 10 am | 11 am |
|---|---|---|---|---|
| Temperature in °F | 75 | 80 | 81 | 87 |

The formula to convert Fahrenheit (F) into Celsius (C) is

$$C = \frac{5}{9}(F - 32)$$

What time was the first check where the water temperature was above 26°C?
You **must** show your working.

**[3 marks]**

8am $C = \frac{5}{9}(75 - 32) = \frac{5}{9}(43) = 23.888\ldots < 26$

9am $C = \frac{5}{9}(80 - 32) = \frac{5}{9}(48) = 26.66\ldots > 26$

Answer 9am

Do not write outside the box

**8 (b)** There are 90 people in the pool.

number of adults : number of children = 3 : 2

One lifeguard is needed for every 10 children.

How many lifeguards are needed for the children in the pool?

**[4 marks]**

$90 \div (3+2) = 18$

$18 \times 2 = 36$ children present

$36 \div 10 = 3.6$

so 4 lifeguards are needed

Answer 4

7

**Turn over for the next question**

Do not write outside the box

**9 Hedgehogs**

Aahad works for a charity that rescues hedgehogs.

**9 (a)** The charity has four main costs each month.

| | Cost per month |
|---|---|
| **Straw** | £25 |
| **Feed** | £39 |
| **Electricity** | £68 |
| **Vets** | £235 |

From next month the costs are changing.

- Straw will be free from a local farm
- Feed is increasing by $\frac{1}{6}$
- Electricity is increasing by £324 a year
- The cost of the vets will be £3090 a year.

The charity receives a new monthly donation of £30

Is the new donation enough to pay for the **increase** in total monthly costs?
You **must** show your working.

**[5 marks]**

feed 39 x 1/6 = 6.5 increase

electricity 324 ÷ 12 = 27 increase

vets 3090 ÷ 12 − 235 = 22.5 increase

straw 25 decrease

6.5 + 27 + 22.5 − 25 = 31 > 30

no the extra donation won't cover the increase in ~~both~~ monthly costs.

Do not write outside the box

**9 (b)** The charity needs a new building for the hedgehogs.

The new building will cost £250 000

The charity invested £245 000 in a bank account 2 years ago.

The account pays 1.2% compound interest each year.

Is there enough money in the bank account to pay for the building?
You **must** show your working.

**[3 marks]**

1.2% 1.012 multiplier

$245000 \times 1.012^2 = 250\,915.28$

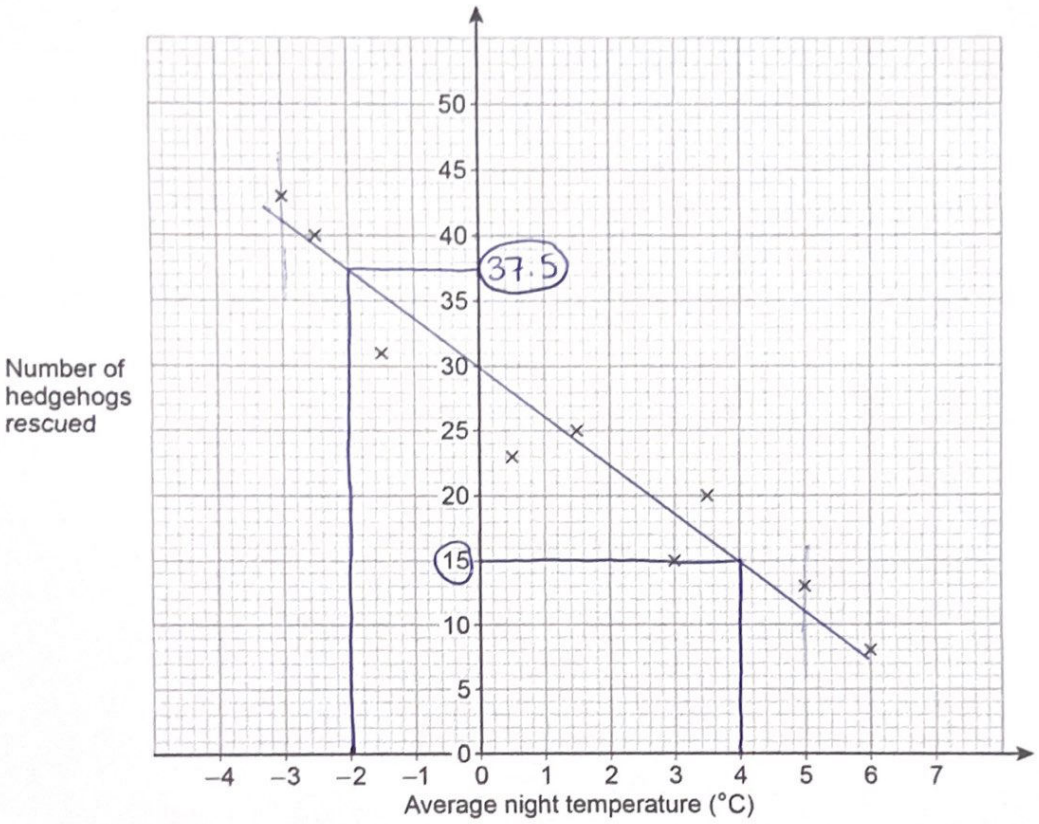
so yes there is enough money in the account to pay for the building.

**Question 9 continues on the next page**

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Each week Aahad records the number of hedgehogs rescued and the average night temperature that week.

The scatter diagram shows information for the last 9 weeks.




**9 (c)** Describe the relationship between average night temperature and number of hedgehogs rescued.

**[1 mark]**

As temperature increases the number of hedgehogs rescued decreases

Do not write outside the box

**9 (d)** In week A the average night temperature was –2°C

In week B the average night temperature was 4°C

Use the scatter diagram to estimate how many **more** hedgehogs were rescued in week A than in week B.

You **must** show your working, some of which should be on the diagram.

**[4 marks]**

-2°C 37.5 on average

4°C 15 on average

37.5 - 15 = 22.5

dependent on your line of best fit.

Answer 22.5

13

**Turn over for the next question**

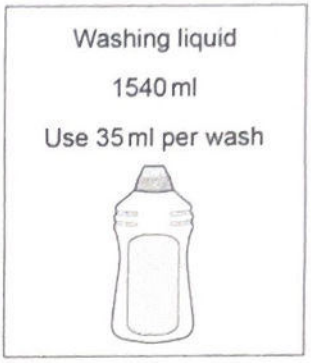
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**10** **Housework**

David is doing housework.

**10 (a)** David is washing clothes.




David says,

"I can do **at least** 32% more washes with the powder than the liquid."

Is he correct?
You **must** show your working.

**[5 marks]**

WP: $3900 \div 65 = 60$ washes

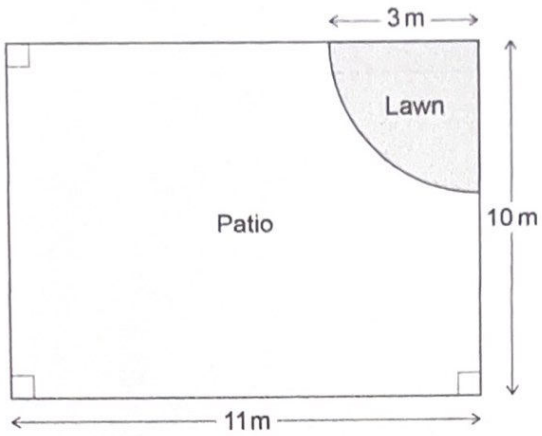
WL: $1540 \div 35 = 44$ washes

$\frac{60 - 44}{44} \times 100 = 36\ \%$

$36\ \% > 32\ \%$ so David is right.

Do not write outside the box

**10 (b)** David is going to clean the patio.




Not drawn accurately

The lawn is a quarter circle with radius 3 m

Each bottle of cleaner

cleans 15 $m^2$ of patio

costs £1.95

Work out the cost of the bottles of cleaner that David needs to clean the patio.
You **must** show your working.

**[6 marks]**

$11 \times 10 = 110\ m^2$ total area

$\frac{1}{4} \pi \times 3^2 = \frac{9}{4} \pi\ m^2$ area of a lawn

$110 - \frac{9}{4} \pi = 102.93\ m^2$ area of patio

$102.93 \div 15 = 6.86$ need 7 bottles

$7 \times 1.95 = 13.65$

Answer £ 13.65

**Question 10 continues on the next page**

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MFS Step IS: Accredited AQA Method & Mark Scheme Verified by Tutor Abdul [M1 A1] [100% PASS]



Do not write outside the box

**10 (c)** David makes a cleaning spray using vinegar and water.

The cleaning spray needs to be

700 ml in total

45% vinegar.

David has put **half a pint** of vinegar in the spray bottle.

How much **more** vinegar and how much water does he need to add?
Use 1 pint = 568 ml
State the units of your answers.

**[6 marks]**

$0.45 \times 700 = 315$ ml of vinegar

$0.5 \times 568 = 284$ ml of vinegar already in

$315 - 284 = 31$ ml more vinegar

$700 - 315 = 385$ ml of water

Vinegar 31

Water 385

17

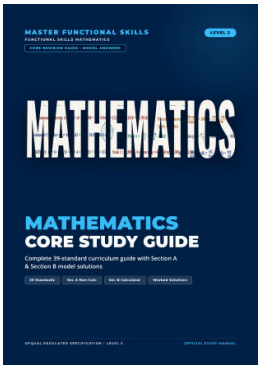
**END OF QUESTIONS**

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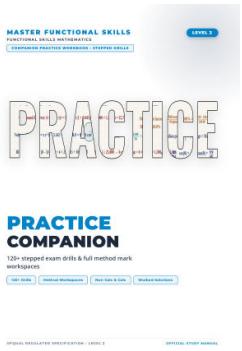



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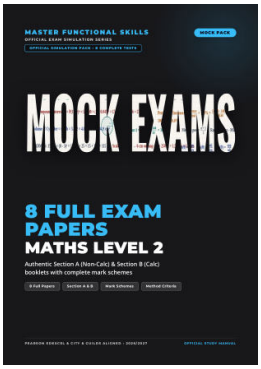



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